

Definition and first terms

$3 A(x) = 2 + 1 x + A(x)^2$, with the origin-normalized solution.
Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+1*T^2$.
 $a(0), \dots, a(7) = 1, 1, 1, 2, 5, 14, 42, 132$.
Kernel used throughout: $\rho(u) = u(1 - 1 u^1)$, so $x = \rho(T(x))$.

Typogeometry



literal 2-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon 1$.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_1) \in \mathbb{Z}_{\geq 0}^1 : \sum_{j=1}^1 j k_j = n - 1\}, \quad K = \sum_{j=1}^1 k_j,$$
$$a(n) = \frac{1}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_1} 1^{k_1}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^1 P_r(n) a(n+r) = 0, \quad n \geq 1, \quad \text{with}$$
$$P_0(n) = -4 n + 2$$
$$P_1(n) = n + 1$$

Differential equation

$$(2 x) A(x) + (-4 x^2 + x) A'(x) = 3 x.$$

Matrices, telescope certificate, and checks

$$G = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 2 & -1 \\ -1 & 1 & 0 & 2 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 4 & 2 \\ 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Certificate.

$$R(n, u) = \frac{2 u - 1}{\rho(u)^0}.$$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 4×4 ; remainder 1×2 , rank 1, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.